



2011 Half-Yearly Examination

FORM V

MATHEMATICS 2 UNIT

Wednesday 11th May 2011

General Instructions

- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question.
- Start each question on a new leaflet.

Structure of the paper

- Total marks — 90
- All six questions may be attempted.
- All six questions are of equal value.

Collection

- Write your name, class and master clearly on each leaflet.
- Hand in the six questions in a single well-ordered pile.
- Hand in a leaflet for each question, even if it has not been attempted.
- If you use a second leaflet for a question, place it inside the first.
- Write your name on the question paper and place it inside your leaflet for Question One.

5P: SJG

5Q: TCW

Checklist

- Writing leaflets: 6 per boy.
- Candidature — 31 boys

Examiner
SJG

QUESTION ONE (15 marks) Start a new leaflet.

(a) Write as a fraction in lowest terms:

(i) 0.12

(ii) 17.5%

(b) Express $\frac{6}{\sqrt{3}}$ with a rational denominator and simplify.

(c) Evaluate $\sin 33^\circ 45'$ correct to two decimal places.

(d) Simplify $4\sqrt{2} + 3 - \sqrt{2}$.

(e) Factorise:

(i) $x^2 - 5x - 36$

(ii) $a^3 + 8$

(f) Determine whether the point $(3, -2)$ lies on the line $2x + 3y - 1 = 0$.

(g) Solve:

(i) $x^2 - 4 = 0$

(ii) $|x| = 7$

(h) Simplify $\frac{8ab}{5} \div \frac{4ab}{15}$.

(i) Solve the inequation $7x - 9 \leq 26$. Graph the solution on a number line.

QUESTION TWO (15 marks) Start a new leaflet.

- (a) Simplify $3\sqrt{45}$.
- (b) (i) On a set of axes, draw a ray representing the angle $\theta = 210^\circ$.
(ii) Find $\cos 210^\circ$.
- (c) How far and in what direction has the parabola $y = x^2$ been translated to produce $y = (x + 4)^2$?
- (d) (i) Solve the inequation $|x + 3| \leq 5$.
(ii) Graph your solution on a number line.
- (e) If $P(x) = x^2 - 3x + 2$, find and simplify:
(i) $P(3)$
(ii) $P(a) + 3$
(iii) $P(a + 3)$
- (f) Simplify $\frac{1}{x} - \frac{1}{x + 1}$.
- (g) Expand and simplify $(\sqrt{3} + 1)^2$.
- (h) Find the acute angle θ to the nearest degree if:
(i) $\tan \theta = 1.4$
(ii) $\operatorname{cosec} \theta = 1.3$

QUESTION THREE (15 marks) Start a new leaflet.

(a) Factorise $3b^3 - 3c^3$.

(b) Find the natural domain of each function:

(i) $f(x) = \frac{1}{x+1}$

(ii) $f(x) = \sqrt{7-x}$

(c) Consider the parabola $y = x^2 + 2x - 8$.

(i) Find the y -intercept.

(ii) Find the x -intercepts.

(iii) Find the coordinates of the vertex.

(iv) Clearly sketch the parabola, showing all intercepts and the vertex.

(v) Hence solve $x^2 + 2x - 8 > 0$.

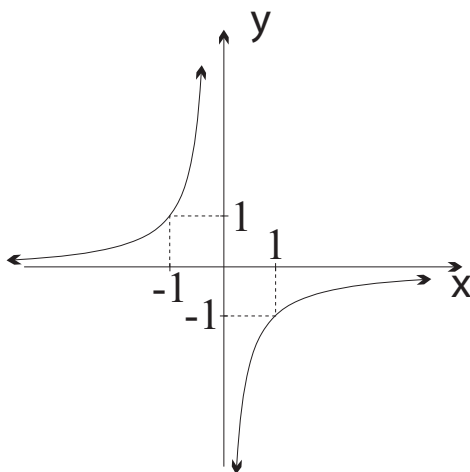
(d) Simplify:

(i) $\sqrt{27} + \sqrt{75} - \sqrt{48}$

(ii) $(2\sqrt{6} + 3)(2\sqrt{6} - 3)$

(e) Solve $\sin x = \frac{1}{\sqrt{2}}$, for $0^\circ \leq x \leq 360^\circ$.

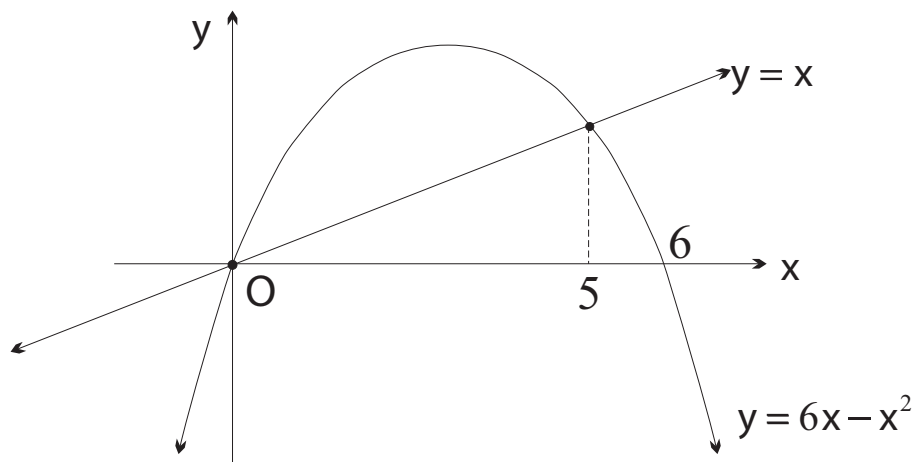
(f)



The graph above shows the curve $y = -\frac{1}{x}$. What is the equation of the vertical asymptote?

QUESTION FOUR (15 marks) Start a new leaflet.

- (a) Two positive numbers differ by 5 and their squares add to 233. Form an equation and solve it to find the two numbers.
- (b) A man standing on a rooftop looks down at a car parked some distance away. If the angle of depression of his line of sight is 18° and he is 16 metres above the ground, how far away from the base of the building is the car? Give your answer to the nearest centimetre.
- (c) Factorise $3x^2 - 10x - 8$.
- (d) Express the following fractions with a rational denominator:
- $\frac{\sqrt{2}}{\sqrt{3}}$
 - $\frac{1}{\sqrt{5} - \sqrt{3}}$
 - $\frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}}$
- (e) Given the function $f(x) = x^3 - x$, find $f(-x)$ and hence determine whether the symmetry of $f(x)$ is even, odd or neither.
- (f)



Using the graph above, or otherwise, solve the inequation $6x - x^2 < x$.

- (g) If a ranger can see his camp on a bearing of 127°T , what is the bearing of the ranger from the camp?

QUESTION FIVE (15 marks) Start a new leaflet.

(a) Showing all working, write $1.\dot{6}0\dot{3}$ as a fraction in lowest terms.

(b) Simplify $\frac{x+1}{x^2-6x+5} - \frac{x-4}{x^2+4x-5} + \frac{1}{x^2-25}$.

(c) If $\cos \theta = \frac{\sqrt{5}}{3}$ and $\tan \theta < 0$, find the exact value of $\sin \theta$.

(d) (i) Find the centre and radius of the circle $x^2 - 4x + y^2 + 2y = 4$.

(ii) Using a graph, or otherwise, determine the range of $x^2 - 4x + y^2 + 2y = 4$.

(e) (i) On a single set of axes, sketch $y = |x - 3|$ and $y = |x + 1|$.

(ii) Hence, or otherwise, solve $|x - 3| = |x + 1|$.

(f) Prove the identity $\cos \theta + \tan \theta \sin \theta = \sec \theta$.

QUESTION SIX (15 marks) Start a new leaflet.

- (a) Sketch the graph of the function $y = \frac{1}{x+2} - 1$, clearly showing any asymptotes and intercepts with the axes.
- (b) Shade the region where $x^2 + y^2 \leq 16$, $x > -1$ and $y < 2$.
- (c) A ship at sea sees a lighthouse on a bearing of 320°T . After the ship sails into port, which lies 6 kilometres due west of its original position, the lighthouse is still visible but on a bearing of 048°T . How far is the port from the lighthouse? Leave your solution correct to 2 decimal places.
- (d) Solve $\cot 2\alpha = -1$, for $0^\circ \leq \alpha \leq 360^\circ$, giving your solutions correct to one decimal place.
- (e) (i) Sketch $y = -\sqrt{16 - x^2}$.
- (ii) Sketch $y = 4 - x^2$, for $-4 \leq x \leq 4$, on the same number plane as part (i).
- (iii) Solve $4 - x^2 = -\sqrt{16 - x^2}$, leaving your solutions as exact values.
- (iv) Hence, or otherwise, solve $4 - x^2 + \sqrt{16 - x^2} \geq 0$.

END OF EXAMINATION

Q1. a) i) $0.12 = \frac{3}{25}$ ✓

ii) $17.5\% = \frac{7}{40}$ ✓

b) $\frac{6}{\sqrt{3}} = 2\sqrt{3}$ ✓

c) $\sin 33^\circ 45' = 0.56$ (to 2 d.p.) ✓

d) $4\sqrt{2} + 3 - \sqrt{2} = 3 + 3\sqrt{2}$ ✓

e) i) $x^2 - 5x - 36 = (x-9)(x+4)$ ✓

ii) $a^3 + 8 = (a+2)(a^2 - 2a + 4)$ ✓

f) $2 \times 3 + 3 \times (-2) - 1 \neq 0$

$\therefore (3, -2)$ is not on the line $2x + 3y - 1 = 0$ ✓

g) i) $x^2 - 4 = 0$

$(x+2)(x-2) = 0$

$x = 2$ ✓ or $x = -2$ ✓

ii) $|x| = 7$

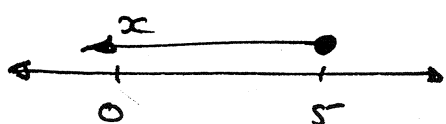
$\therefore x = 7$ ✓ or $x = -7$ ✓

h) $\frac{8ab}{5} \div \frac{4ab}{15} = \frac{8ab}{5} \times \frac{15}{4ab}$
 $= 6$ ✓

i) $7x - 9 \leq 26$

$7x \leq 35$

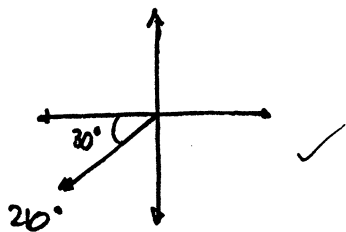
$x \leq 5$ ✓



15

Q2. a) $3\sqrt{45} = 9\sqrt{5}$ ✓

b) i)



ii) $\cos 210^\circ = -\cos 30^\circ$
 $= -\frac{\sqrt{3}}{2}$ ✓

c) shifted left 4 units ✓

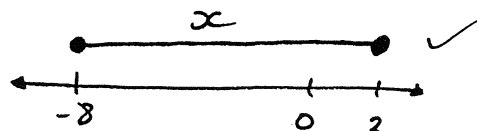
d) i) $|x+3| \leq 5$

$x+3 \leq 5$ and $-x-3 \leq 5$

$x \leq 2$ and $-x \leq 8$ ✓ one answer only
 $x \geq -8$

$\therefore -8 \leq x \leq 2$ ✓ both solutions and interval

ii)



e) i) $P(3) = 9 - 9 + 2$
 $= 2$ ✓

ii) $P(a)+3 = a^2 - 3a + 2 + 3$
 $= a^2 - 3a + 5$ ✓

iii) $P(a+3) = (a+3)^2 - 3(a+3) + 2$ ✓
 $= a^2 + 6a + 9 - 3a - 9 + 2$
 $= a^2 + 3a + 2$ ✓

f) $\frac{1}{x} - \frac{1}{x+1} = \frac{x+1-x}{x(x+1)}$
 $= \frac{1}{x(x+1)}$ ✓

g) $(\sqrt{3}+1)^2 = 3 + 2\sqrt{3} + 1$
 $= 4 + 2\sqrt{3}$ ✓

h) i) $\tan^{-1}(1.4) = 54^\circ$ (to nearest degree) ✓

ii) $\operatorname{cosec}^{-1}(1.3) = \frac{1}{\sin^{-1}(1.3)}$

$= 50^\circ$ (to nearest degree) ✓

Q3. a) $3b^3 - 3c^3 = 3(b-c)(b^2+bc+c^2)$ ✓

b) i) $x+1 \neq 0$

$\therefore x \neq -1$ ✓

ii) $7-x \geq 0$

$x \leq 7$ ✓

c) i) y-intercept: $y = -8$ ✓

ii) x-intercepts: $x^2 + 2x - 8 = 0$

$(x+4)(x-2) = 0$ ✓

$\therefore x = -4$ and $x = 2$ ✓

iii) vertex: $x = \frac{-b}{2a}$

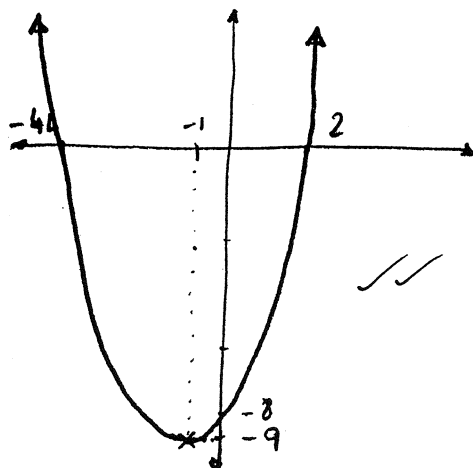
$= -1$

$\therefore y = (-1)^2 + 2(-1) - 8$

$= -9$

$\therefore$ vertex is $(-1, -9)$ ✓

iv)



v) $x^2 + 2x - 8 > 0$

$\therefore x < -4$ or $x > 2$ ✓

d) i) $\sqrt{27} + \sqrt{75} - \sqrt{48} = 3\sqrt{3} + 5\sqrt{3} - 4\sqrt{3}$

$= 4\sqrt{3}$ ✓

ii) $(2\sqrt{6} + 3)(2\sqrt{6} - 3) = 24 - 9$

$= 15$ ✓

e) $\sin x = \frac{1}{\sqrt{2}} \therefore x = 45^\circ$ or $x = 135^\circ$ ✓

f) $x = 0$ ✓

Q4. a) Let the smaller number be x .

$$x^2 + (x+5)^2 = 233 \quad \checkmark$$

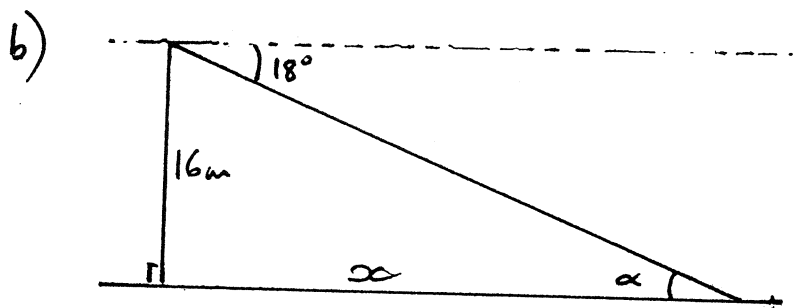
$$x^2 + x^2 + 10x + 25 = 233$$

$$2x^2 + 10x - 208 = 0$$

$$(x+13)(2x-16) = 0 \quad \checkmark$$

$\therefore x = 8$ or $x = -13$ (not a solution to the problem)

Hence the two numbers are 8 and 13. $\checkmark$



$\alpha = 18^\circ$ (alternate angles on parallel lines)

$$\tan 18^\circ = \frac{16}{x} \quad \checkmark$$

$$x = \frac{16}{\tan 18^\circ} \\ = 49.24 \text{ m} \quad \checkmark$$

$$\begin{aligned} \text{c) } 3x^2 - 10x - 8 &= 3x^2 - 12x + 2x - 8 \quad \checkmark \\ &= 3x(x-4) + 2(x-4) \\ &= (3x+2)(x-4) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{d) i) } \frac{\sqrt{2}}{\sqrt{3}} &= \frac{\sqrt{6}}{3} \quad \checkmark & \text{ii) } \frac{1}{\sqrt{5}-\sqrt{3}} &= \frac{\sqrt{5}+\sqrt{3}}{5-3} \\ & & &= \frac{\sqrt{5}+\sqrt{3}}{2} \quad \checkmark & \text{iii) } \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} &= \frac{(\sqrt{3}+\sqrt{2})^2}{3-2} \\ & & & & &= 3+2\sqrt{6}+2 \quad \checkmark \\ & & & & &= 5+2\sqrt{6} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{e) } f(-x) &= (-x)^3 - (-x) \\ &= -x^3 + x \quad \checkmark & \therefore f(x) &= -f(-x) \text{ and } f(x) \text{ is } \underline{\text{odd}}. \quad \checkmark \end{aligned}$$

f) from the graph: $6x - x^2 < x$ when $x < 0$ or $x > 5$. $\checkmark$

g) Bearing of ranger from the camp: $127 + 180 = 307^\circ \text{ T} \quad \checkmark$

Q5. a) $x = 1.\dot{6}0\dot{3}$

$1000x = 1603.\dot{6}0\dot{3} \quad \checkmark$

$\therefore 999x = 1602$

$x = \frac{178}{111} \quad \checkmark$

b)
$$\begin{aligned} \frac{x+1}{x^2-6x+5} - \frac{x-4}{x^2+6x-5} + \frac{1}{x^2-25} &= \frac{x+1}{(x-5)(x-1)} - \frac{x-4}{(x+5)(x-1)} + \frac{1}{(x+5)(x-5)} \quad \checkmark \\ &= \frac{(x+1)(x+5) - (x-5)(x-4) + (x-1)}{(x+5)(x-5)(x-1)} \quad \checkmark \\ &= \frac{x^2+6x+5 - x^2+9x-20 + x-1}{(x+5)(x-5)(x-1)} \\ &= \frac{16x-16}{(x+5)(x-5)(x-1)} \\ &= \frac{16}{(x+5)(x-5)} \quad \checkmark \end{aligned}$$

c) $\cos \theta = \frac{\sqrt{5}}{3} \quad \therefore \cos^2 \theta = \frac{5}{9}$

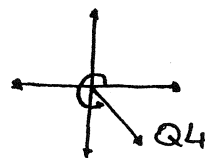
$\therefore \sin^2 \theta = \frac{4}{9} \quad \checkmark$

$\sin \theta = \pm \frac{2}{3}$

$\tan \theta < 0$

$\therefore \sin \theta < 0$

$\therefore \sin \theta = -\frac{2}{3} \quad \checkmark$



d) i) $x^2 - 4x + y^2 + 2y = 4$

$(x-2)^2 + (y+1)^2 = 9$

$\therefore$ circle of centre $(2, -1) \quad \checkmark$

and radius $= 3 \quad \checkmark$

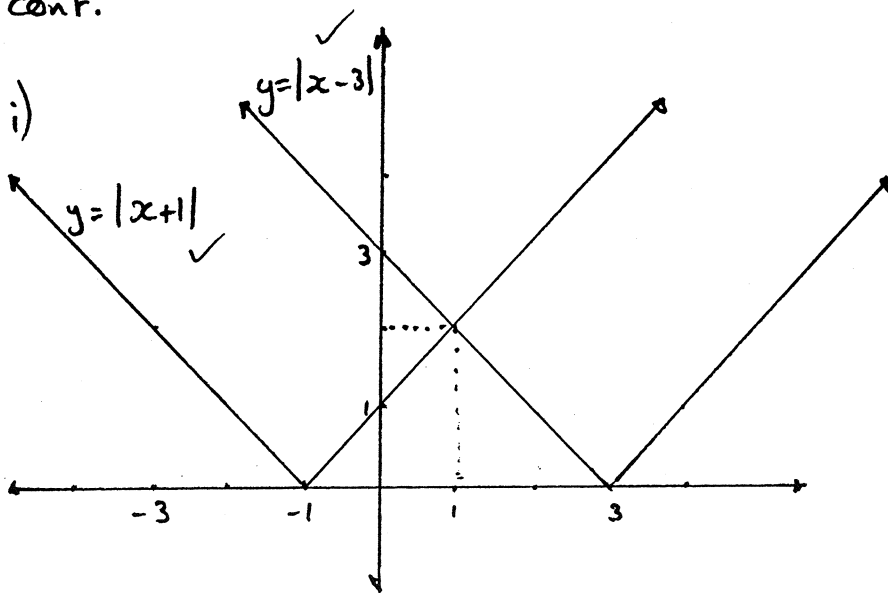
ii) range is centre $\pm$ radius

$\therefore -4 \leq y \leq 2 \quad \checkmark$

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Q5. cont.

e) i)



ii) intersection: $|x-3| = |x+1|$
when $x=1$ ✓

f) LHS: $\cos \theta + \tan \theta \sin \theta = \cos \theta + \frac{\sin^2 \theta}{\cos \theta}$ ✓

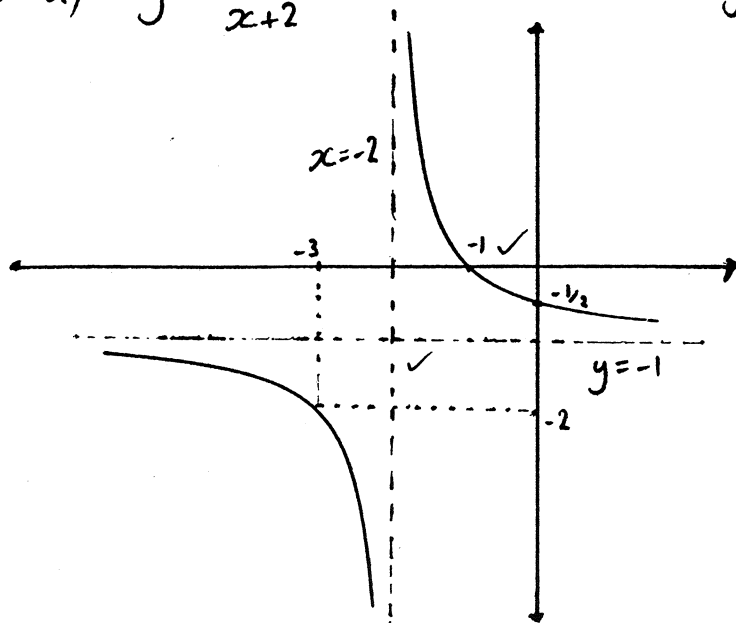
$$= \frac{\cos^2 \theta}{\cos \theta} + \frac{\sin^2 \theta}{\cos \theta}$$

$$= \frac{1}{\cos \theta}$$

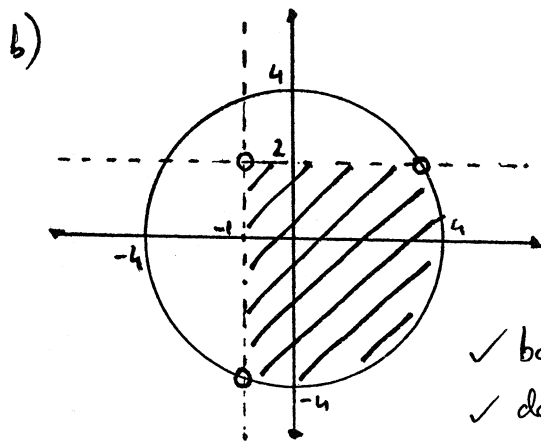
$$= \text{RHS} \quad \checkmark$$

/15

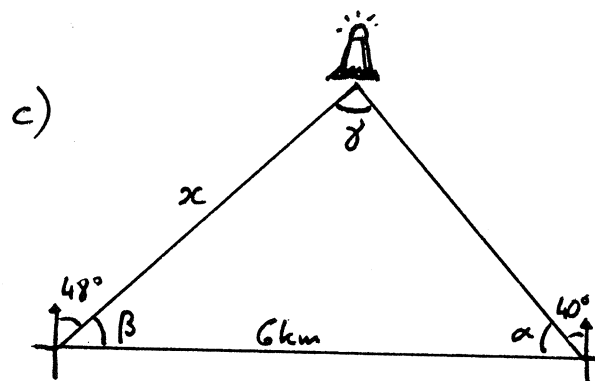
Q6. a) $y = \frac{1}{x+2} - 1$ is the curve $y = \frac{1}{x}$ shifted left 2 units and down 1 unit.



✓ hyperbola



✓ boundaries
✓ dotted/full
✓ lines/curves/
shading



$\alpha = 50^\circ$ (adjacent angles)

$\beta = 42^\circ$ (adjacent angles)

$\therefore \gamma = 88^\circ$ (angle sum of triangle)

$\therefore \frac{x}{\sin \alpha} = \frac{6}{\sin \gamma} \quad \therefore x = \frac{6 \sin 50^\circ}{\sin 88^\circ}$

$= 4.60 \text{ km (2 d.p.)}$

d) $\cot 2\alpha = -1 \quad \therefore \text{let } u = 2\alpha$

$\therefore \cot u = -1, \quad 0^\circ \leq u < 720^\circ$

$\tan u = -1$ (Q2 and Q4) ✓

$u = 135^\circ$

$\alpha = 67.5^\circ$

or $u = 315^\circ$

$\therefore$ or $\alpha = 157.5^\circ$

or $u = 495^\circ$

or $\alpha = 247.5^\circ$

or $u = 675^\circ$

or $\alpha = 337.5^\circ$

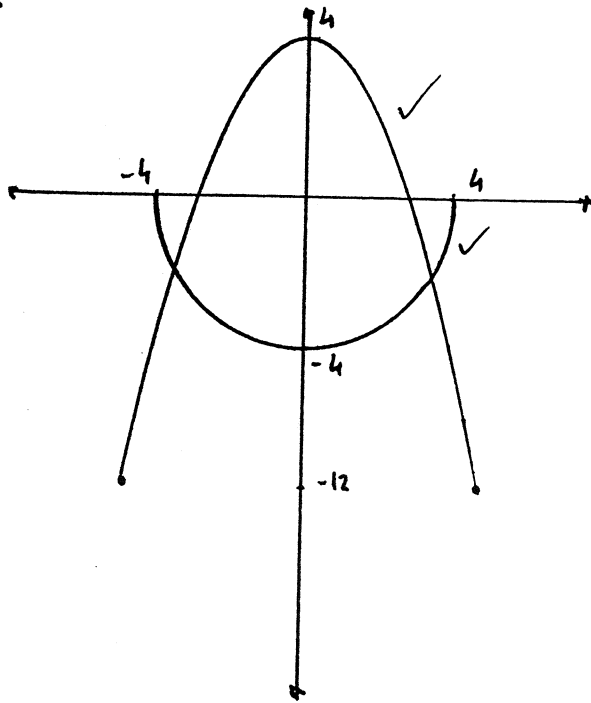
✓ 4 solutions

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Q6. cont.

e) i)

ii)



$$\text{iii) } (4 - x^2) = -\sqrt{16 - x^2}$$

$$(4 - x^2)^2 = 16 - x^2$$

$$16 - 8x^2 + x^4 = 16 - x^2$$

$$x^4 - 7x^2 = 0$$

$$x^2(x^2 - 7) = 0$$

$$\therefore x^2 = 0 \quad \text{or} \quad x^2 = 7$$

$$x = 0$$

$$x = \pm \sqrt{7} \quad \checkmark$$

not a solution $\checkmark$
(see graph)

two solutions
only

$$\text{iv) from graph: } -\sqrt{7} \leq x \leq \sqrt{7} \quad \checkmark$$